Performance Evaluation: Extracting Meaningful Insights

Once the backtest is complete, we need to evaluate the results to understand how well the strategy performed and identify potential improvements. Professional traders look beyond simple profit and loss to analyze various performance metrics:

```
def evaluate_backtest(backtest_results):
    """
    Evaluate the results of a backtest with comprehensive metrics

    Parameters:
    backtest_results (pd.DataFrame): DataFrame with backtest results

    Returns:
    dict: Dictionary of performance metrics
    """
    # Extract relevant data
    portfolio_values = backtest_results['Portfolio Value']
    initial_capital = portfolio_values.iloc[0]
    final_capital = portfolio_values.iloc[-1]

    # Calculate daily returns
    backtest_results['Daily Return'] = portfolio_values.pct_change()
    returns = backtest_results['Daily Return'].dropna()

    # Calculate basic performance metrics
    total_days = len(returns)
    trading_days_per_year = 252
    years = total_days / trading_days_per_year

    # Total return
    total_return = (final_capital / initial_capital) - 1

    # Annualized return
    annualized_return = (1 + total_return) ** (1 / years) - 1

    # Volatility (annualized standard deviation of returns)
    volatility = returns.std() * np.sqrt(trading_days_per_year)
```